

# SSH Honeypot & Attack Log Analysis Lab

Incident Analysis Report | Academic / Portfolio Project

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## 1. Project Overview

This project deployed and tested an SSH honeypot using Cowrie in an isolated VMware laboratory. Ubuntu hosted the honeypot and Kali Linux generated controlled SSH activity. The objective was to collect security telemetry and practice a basic SOC-style investigation from connection through authentication and post-authentication activity.

## 2. Lab Environment

| Component           | Value                        |
|---------------------|------------------------------|
| Test client         | Kali Linux - 192.168.216.131 |
| Honeypot            | Ubuntu - 192.168.216.133     |
| Honeypot technology | Cowrie 3.0.13                |
| Protocol            | SSH                          |
| Listening port      | TCP/2222                     |
| Virtualization      | VMware                       |

## 3. Deployment & Validation

- Created a dedicated non-root cowrie user and isolated Python virtual environment.
- Installed Cowrie 3.0.13 and initialized configuration and logging directories.
- Started Cowrie and verified the service state and TCP/2222 listener.
- Connected from Kali to the honeypot and generated controlled test authentication and shell activity.
- Collected connection, authentication, SSH client, HASSH fingerprint, command and TTY-session evidence.

## 4. Evidence & Timeline

| Time (UTC) | Event          | Source          | Observation                                             |
|------------|----------------|-----------------|---------------------------------------------------------|
| 17:43:14   | SSH connection | 192.168.216.131 | Connection to 192.168.216.133:2222                      |
| 17:44:14   | SSH connection | 192.168.216.131 | Second connection established                           |
| 17:44:54   | Authentication | 192.168.216.131 | testuser / training123 - simulated honeypot success     |
| 23:06:16   | SSH connection | 192.168.216.131 | New session created                                     |
| 23:06:45   | Authentication | 192.168.216.131 | fakeuser / wrongpass1 - simulated honeypot success      |
| 23:07:15   | Command        | 192.168.216.131 | exit                                                    |
| 23:07:19   | SSH connection | 192.168.216.131 | New session created                                     |
| 23:07:29   | Authentication | 192.168.216.131 | admin-test / wrongpass2 - simulated honeypot success    |
| 23:07:44   | Command        | 192.168.216.131 | Shell activity recorded                                 |
| 23:07:54   | Command        | 192.168.216.131 | exit                                                    |
| 23:08:01   | SSH connection | 192.168.216.131 | New session created                                     |
| 23:08:07   | Authentication | 192.168.216.131 | security-test / wrongpass3 - simulated honeypot success |
| 23:18:37   | Command        | 192.168.216.131 | whoami                                                  |
| 23:18:46   | Command        | 192.168.216.131 | pwd                                                     |
| 23:18:56   | Command        | 192.168.216.131 | uname -a                                                |
| 23:19:05   | Command        | 192.168.216.131 | ls                                                      |
| 23:19:10   | Command        | 192.168.216.131 | ls                                                      |

## 5. Key Indicators & Telemetry

| Item                  | Observed value                    |
|-----------------------|-----------------------------------|
| Source IP             | 192.168.216.131                   |
| Target                | 192.168.216.133:2222              |
| SSH client            | OpenSSH_10.3p1 Debian-4           |
| HASSH fingerprint     | eecca2460550b9ded084ecf2f70a75356 |
| Authentication events | 4 controlled simulated successes  |
| TTY evidence          | 4 recorded session files          |

## 6. Analysis

The same lab source IP generated multiple SSH connections and authentication events against the honeypot. Three controlled authentication attempts occurred within approximately two minutes, followed by recorded shell-session activity. The repeated source IP and repeated HASSH fingerprint provide simple correlation points for a SOC analyst. The later session demonstrated observable post-authentication commands, which can be used to distinguish authentication from subsequent activity.

The word “succeeded” in Cowrie authentication logs should not be interpreted as compromise of a real system account. Cowrie is an emulated honeypot and intentionally accepts simulated credentials to create a realistic session for monitoring and analysis.

## 7. Detection Scenario

Detection hypothesis: alert when multiple SSH authentication events originate from the same source IP against the same host within a short time window, then correlate the source with the SSH client fingerprint and post-authentication commands. A production SOC would enrich this alert with asset criticality, geolocation, allowlists, historical activity and other telemetry.

## 8. Recommended Defensive Controls

- Restrict SSH exposure to trusted networks and administrative paths.
- Prefer key-based authentication and minimize password-based access.
- Centralize authentication and system logs for correlation and retention.
- Alert on repeated authentication activity and unusual post-authentication commands.
- Apply firewall policies and network segmentation.
- Review SSH configuration, account controls and privileged access regularly.

## 9. Limitations & Scope

All activity was generated in the student's own VMware laboratory. The IP addresses are private lab addresses, not public indicators of compromise. The project demonstrates a beginner SOC investigation workflow and does not represent production incident response or real-world malicious activity.

## 10. Evidence for Portfolio

- Cowrie status and PID confirmation.
- TCP/2222 listening confirmation using ss.
- Kali SSH connection to the honeypot.
- Authentication and connection log screenshots.
- Command log screenshots.
- TTY/session file evidence.
- Final investigation timeline and recommendations.